Project 2: Research Study

CS 5090 · Software Engineering with Generative AI

Solo or in pairs · 45% of course grade · Final paper due Fri, Dec 11

1. Goal

The goal of Project 2 is to carry out a small research study on generative AI in software engineering. You will define one focused research question, design an evaluation, run the study, and report the results in a short conference-style paper. The final paper may be up to 8 pages, excluding references.

You may work alone or with one partner. Pairs should take on a somewhat larger scope, but the grading criteria are the same. The proposal is where we will agree on a reasonable scope for your project.

Strong projects may be suitable for submission to a workshop or similar venue. If the instructor approves the paper for submission and the team submits it after the course, the project receives a +5% bonus.

2. Milestones and Grading

Date	Milestone	What you submit	Weight
Fri, Sep 18	Topics circulated	Candidate topics and possible directions posted	—
Fri, Oct 2	Research abstract	150–250 words: problem, gap, proposed study, expected result	5%
Oct 19 – Nov 4	Related work + pitch	10–20 minutes + 1–2 minutes Q&A, in a class you sign up for: 1–2 related-work papers, then your project pitch (question, motivation, evaluation plan)	5%
Sun, Nov 8	Proposal	3–5 pages: related-work table, method, baseline, metrics, week-by-week plan; pilot result recommended	15%
Fri, Dec 4	Full draft	Complete draft with results available at that point	15%
Dec 7 / Dec 9	Presentation	8 minutes + 2 minutes Q&A	15%
Fri, Dec 11	Final submission	Final paper + artifact (code, data, scripts, prompts/configs as appropriate)	45%

Feedback: You will receive written feedback on the early milestones. Use it to revise the project as it develops. You are not required to follow every suggestion, but if you choose a different direction, be prepared to explain why. The instructor may also provide feedback during the project as needed.

3. Choosing a Research Question

You may develop the project through any of the following routes:

Start from a paper. Replicate, extend, or challenge a result from a paper in software engineering and AI. Examples include testing another model, dataset, setting, or failure condition. See suggested pool of SE papers.

Enter a competition. Choose a task from the ICSE 2027 Competition Track. The task provides the dataset and evaluation setup; your paper explains your approach and results.

Use the AIDev dataset. Study an empirical question about how AI coding agents contribute to GitHub pull requests using the AIDev dataset.

Based on above routes, please propose your own topic. Any question within the scope of the course is acceptable with instructor approval. A good project question should be specific enough to test during the semester. By the proposal stage, you should be able to state the main claim or hypothesis clearly and describe what evidence would support or contradict it.

4. What Counts as a Research Study

The project should go beyond building a demo. It needs an evaluation that supports a research claim. Depending on the topic, this may involve controlled experiments, benchmark evaluation, dataset analysis, human evaluation, or a combination of these.

At minimum, the final study should include:

- a clear research question or claim;
- a meaningful comparison, baseline, or reference point;
- an evaluation metric that matches the research question;
- enough runs, examples, or observations to support the conclusion;
- model, dataset, prompt, seed, configuration, and software-version details needed for reproducibility, where applicable;
- a discussion of limitations and threats to validity;
- a brief discussion of benchmark contamination if you use a public benchmark for code LLMs.

5. Review Criteria

Your final paper is judged on the criteria a real research track uses. The five below are the review criteria of the ICSE Research Track, reproduced here so that you can check your own paper before submitting it and so that the feedback you receive at each milestone uses a consistent vocabulary.

- **Novelty.** The novelty and innovativeness of contributed solutions, problem formulations, methodologies, theories, and/or evaluations, that is, the extent to which the paper is sufficiently original with respect to the state of the art. In this course, a careful replication that adds a new model, dataset, or condition counts; re-running a published experiment with nothing added does not.
- **Rigor.** The soundness, clarity, and depth of a technical or theoretical contribution, and the level of thoroughness and completeness of an evaluation. In practice: a real baseline, enough runs or observations, reported variance, and honest treatment of results that did not go your way.
- **Relevance.** The significance and/or potential impact of the research on the field of software engineering. Be able to say who would change what they do if your result holds.
- **Verifiability and transparency.** The extent to which the paper includes sufficient information to understand how an innovation works; to understand how data was obtained, analyzed, and interpreted; and how the paper supports independent verification or replication of the claimed contributions. This is what your artifact and README are for, and they will be checked.
- **Presentation.** The clarity of the exposition in the paper.

The paper should clearly explain and justify the claimed contributions. A course project is not expected to score highly on all five; it is expected to be honest about where it stands on each.

6. Suggested Workflow

Weeks 3–5: Abstract. Write a short abstract before committing to the full study. The purpose is to force the problem, gap, and proposed evaluation into a compact form.

Weeks 5–9: Refine the question. Use course readings, Project 1, and early experiments to narrow the scope. Your related-work and pitch talk (Oct 19 – Nov 4) should cover one or two related-work papers and then a concrete question and evaluation plan.

Week 10: Proposal. The proposal should make the study executable: what you will test, on what data, against which baseline, and with which metrics.

Weeks 11–13: Run the study. Carry out the experiments and record enough detail to reproduce them.

Weeks 13–14: Write and revise. Submit a full draft on Dec 4, present on Dec 7 or 9, and use the feedback from both to revise the final paper.

7. Use of Generative AI

Generative AI tools usage follows the course policy. You may use them for coding, experiment setup, data processing, brainstorming, editing, and other research tasks. However, you remain responsible for the correctness of the work.

- Disclose the AI tools you used and what you used them for.
- Verify generated code, analysis, factual claims, and citations.

Appendix A. Research Abstract (due Fri, Oct 2)

Submit 150–250 words. State at the top whether the project is solo or a pair project (and list both names for a pair).

- **Context.** What problem or setting are you studying, and why does it matter?
- **Gap.** What is still unclear based on the papers you have read? Cite the most relevant work.
- **Study.** What will you build, run, compare, or measure?
- **Expected result.** What do you currently expect to find? State the expectation in a way that could turn out to be wrong.

Appendix B. Proposal (due Sun, Nov 8)

Submit 3–5 pages, plus the pilot table if you include one. The proposal can later become the starting point for the introduction and method sections of the final paper.

1. Abstract. Revised version of your Oct 2 abstract.

2. Research question / claim. State the main question or hypothesis clearly.

3. Motivation. Why is the question worth studying? Who would care about the answer?

4. Related work, as a table. List 5–8 relevant papers, one row each, with columns for research question, data or benchmark, metric, main result, and how your project differs. A short discussion may follow the table.

5. Method. Describe the model(s), dataset, task, baseline, variables, prompts/configuration, number of runs, and other experimental details.

6. Evaluation. State the metrics and explain why they answer the research question. Describe what result would weaken or contradict your claim.

7. Feasibility and week-by-week plan. List the compute/resources you need and the main risk or fallback plan, then give a week-by-week plan for Nov 9 – Dec 4: what gets done each week and, for pairs, who owns it.

8. Pilot result (optional, recommended). Run your pipeline end to end on a small slice, for example 10–20 examples with one model and one baseline, and report the numbers in a small table. A pilot that exposes a problem is a good outcome: finding it now is what the proposal is for.

9. Threats to validity. Identify the limitations you already expect.

10. Team plan. For pairs, state who is leading which parts of the work. Both students must contribute to the writing.

Final Submission Checklist

Submit the following by Fri, Dec 11:

- the final paper (up to 8 pages, excluding references);
- the project artifact: code, scripts, data or data-processing instructions, and prompts/configuration files as appropriate;
- a short README explaining how to reproduce the main experiment or analysis;
- an AI-use disclosure in the paper.

The final paper will be evaluated primarily on the clarity of the research question, the quality of the study design, the strength and transparency of the evidence, and how well the conclusions are supported by the results. A negative or unexpected result is completely acceptable if the study is well designed and reported.